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Khmer sandstone quarries of Kulen Mountain and Koh Ker: a petrographic and geochemical study

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ABSTRACT

Sandstone from several ancient quarries and natural outcrops located at the eastern foothill of Kulen Mountain and in Koh Ker, northern Cambodia, has been characterized by means of petrographic analysis, scanning electron microscopy, and geochemical analysis. The samples have been collected during a series of field surveys organized jointly by The Metropolitan Museum of Art and the Authority for the Protection and Management of Angkor and the Region of Siem Reap (APSARA). The data show how the sandstone quarried from these sites is characterized by an overall petrographic and geochemical homogeneity, and is consistent with the upper terms of the Lower–Middle Jurassic Terrain Rouge Formation.

This result suggests that petrography and geochemistry of sandstone belonging to the Terrain Rouge Formation might not be unambiguously characteristic of single quarry sites, but rather representative of vast portions of the sedimentary sequence.

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1. Introduction

Clear traces of organized exploitation of stone material, in form of trenches and step-like surfaces with chisel marks and wedge holes, have been documented all along the eastern edge of Kulen Mountain and, to a lesser extent, on top of the plateau, since the 19th century, at the time of the early archaeological exploration of Cambodia (Delaporte, 1880; Garnier, 1873; Parmantier, 1913) (Fig. 1). These finds consolidated the opinion that Mesozoic sandstone constituting the buttes surrounding Angkor provided building materials for the numerous Khmer monuments (Saurin, 1954; Delvert, 1963). Delvert (1963) provides a detail list of early well-known ancient quarry sites in the area, which includes *Phnom Bei, Trapeang Thmo Dap, Poeng Preah Anglong and O'Mealea*, together with more generic allusion to stone quarries north of Kulen Mountain and in Koh Ker (Delvert, 1963), the major Angkorian political center founded by king Jayavarman IV (928–944) (Jacques and Lafond, 2004). More recently, the quarry of *Trapeang Toek*, and other numerous and mostly undocumented point sources located 5–10 km west of the Angkorian temple of Beng Mealea, in the area of *Don En*, have been reported by other authors (Rocks, 2009). All these sites are constituted of several cuts of shallow depth and with step-like surfaces, scattered on vast areas and apparently independents.

A similar setting is found in Koh Ker, where vast terraces have been exploited, taking advantage of the nearly horizontal sandstone beds. These large outcrops were described by early scholars (Parmentier, 1939), although only one quarry, called *Ang Khna*, was identified at that time. At the beginning of the 21st century, with the progression of the landmine clearance, new sites have been brought to light, proving that monolithic blocks were quarried and dressed locally (Moriai et al., 2002; Evans, 2009).

Recent publications of petrographic and geochemical studies of Angkor sandstones suggest the existence of several local sandstone quarries with different petro-physical properties which can be correlated to different construction phases of Angkor monuments (Uchida et al., 2007). However, their specific provenance is still matter of some debate (Kučera et al., 2008), mostly because of the lack of a systematic characterization of the available stone materials. Whether and how the numerous quarries scattered around Angkor differ in term of the petrographic and geochemical characteristics of the quarried material has not yet been clarified, making considerations on material usage and environment exploitation purely speculative.

This study aims to provide a quantitative characterization of sandstone from some of the numerous pit quarries opened by

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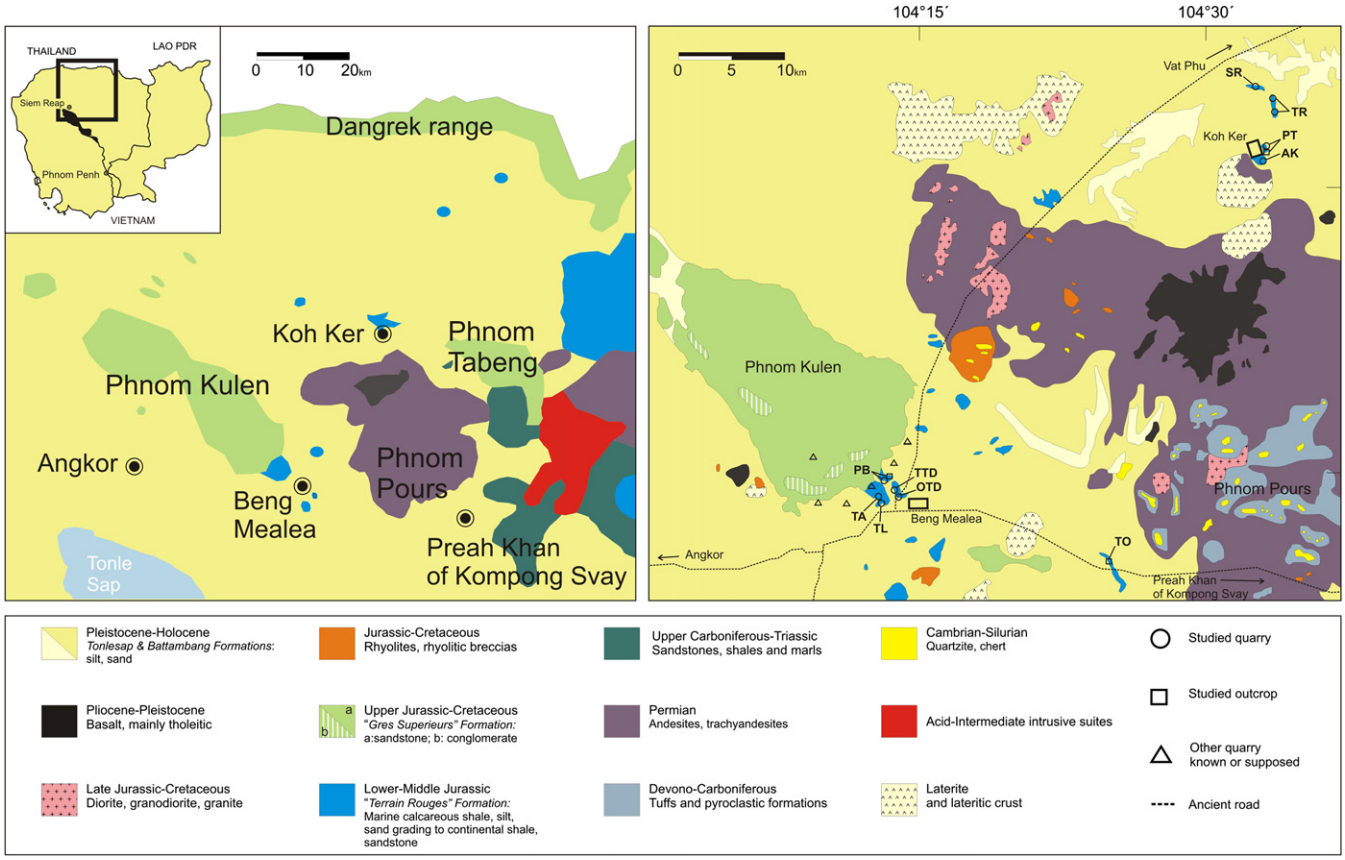


Fig. 1. Schematic geological map of northern Cambodia (left), and of the studied area (right), with superimposed the main temples, the ancient roads (Hendrickson, 2010), and the location of sampled quarries. Geological data modified after Contri, 1972. TL: Toek Lick; TA: Thmo Andong; OTD: O' Thmo Dap; TTD: Trapeang Thmo Dap; PB: Phnom Bei; TO: Ta Ong; AK: Ang Khna; PT: Prasat Thneng; TR: Trapeang Russei; SR: Stung Rongea.

the Khmer during the Angkor period (c. 9th–15th century AD). The studied sandstone originates from both well-known and newly discovered quarry sites in the areas of Beng Mealea and Koh Ker.

2. Materials and methods

2.1. Quarries and sampling campaign

The surveyed quarries are located at the foothill of Kulen Mountain, west to the site of Beng Mealea, and in the surroundings of Koh Ker. Both the areas share a similar setting, with multiple open quarries of variable size and geometry which follow the sandstone attitude (strike and dip of the sandstone beds). Remains of the quarrying activity include many stepped surfaces up to 5 m high and boulders with clear chisel marks, wedge holes, and channels. The variable depth of the pits suggests that the quarries have been abandoned at different stages of exploitation.

At Kulen Mountain five sites were surveyed, known as *Phnom Bei* (three hills), *Trapeang Thmo Dap* (pond of stone cutting), *O' Thmo Dap* (stream of stone cutting), *Thmo Andong* (stone well) and *Toek Lick* (water spring) (Fig. 1). All these sites are located on the western slope of Kulen. The first three occupy a series of low, narrow hills oriented NW–SE and constituting the northern limit of the sandstone outcrop.

The area of *Phnom Bei* occupies the highest topographic position among the studied sites. Here, thinly bedded, poorly consolidated sandstone lays on more compact, massive sandstone with festoon cross-bedding. The quarries, all of shallow depth and mostly filled by soil, are located on the southwest side of the hill. Samples come from the upper, medium and lower portions of the exposed sequence and include documented quarry pits and rock outcrops.

The quarries of *Trapeang Thmo Dap* and *O' Thmo Dap* (Fig. 1) are located at the bottom of the foothills where the sandstone is exposed on washouts and streams. In these sites the pits are shallow and the quarries develop mostly horizontally on compact, massive sandstone. Thin bedding is visible on weathered horizons. At *Trapeang Thmo Dap* samples have been taken from the modern pits opened in the late 1990s for conservation purposes by the Authority for the Protection and Management of Angkor and the Region of Siem Reap (APSARA), and from remains of an ancient, shallow pit quarry nearby, now converted into rice field. In both cases the samples are located close to the topographic surface.

Thmo Andong and *Toek Lick* (Figs. 1 and 2) are the most vast and developed sites in the studied area and can be considered part of an extensive network of quarries, more than isolated and independent points of extraction. Quarries include various pits of different high, clear remains of workshops and unfinished sculptures scattered in an area of about 40 ha. Together, they occupy the southern boundary of the low sandstone outcrop. The sandstone crops out in almost horizontal layers characterized by beds of various thickness and structure, often separated by silty interbeds. Planar- and cross-lamination, festoon cross-lamination and massive beds are visible throughout the site. The sequence, as seen on the quarry walls, is occasionally interrupted by layers rich in reworked carbonate clasts



Fig. 2. Examples from the surveyed quarries at the eastern foothill of Kulen Mountain and in Koh Ker. Clockwise, from upper left: Toek Lick, Thmo Andong, Stung Rongea and Trapeang Russei.

up to few centimeters in size, possibly caliche material (Racey et al., 1996).

The height of the walls, size and sorting of the quarried blocks varies throughout the sites. If some quarries testify an organized exploitation of regular blocks, others in close proximity bear traces of inconsistent block sizing, so that it is arduous to attempt a relative chronology of the quarrying activity.

Where possible, top, middle and bottom of the outcrops have been sampled, although in some cases, mostly for single, shallow outcrops, only one sample from each pit has been taken.

In Koh Ker four main sites have been sampled, all constituted by several pit quarries more or less organized and located at the emergence of massive sandstone terraces or boulders with evidences of planar and festoon cross-lamination: *Ang Khna, Prasat Thneng, Stung Rongea* (the cold stream) and *Trapeang Russei* (pond of bamboo) (Fig. 1).

Ang Khna is located in the southern side of *Trapeang Khna* and it is a well-known site in Koh Ker, as the outcrops have been carved with religious and animistic motifs (Parmentier, 1939). Traces of small quarrying activity are scattered on an area of about 3 ha without a specific organization. No stepped surfaces have been found, but traces of irregular blocks quarried using shallow trenches from the vast boulders which border the Rahal, the main water reservoir of the site (Evans, 2009).

The second quarry is located 1 km east Prasat Thom, on the bed of a small seasonal creek that runs east of *Prasat Thneng*. This narrow outcrop bears the traces of fairly regular exploitation, while piles of stone chips still emerging from the ground could testify the local dressing of the blocks. Additional samples have been collected nearby from the outcrops carved in the shape of linga and housed in three small temples known as G, H and I (Jacques and Lafond, 2004).

The last two sites are located about 6 km N and NE of Prasat Thom.

The quarry of *Stung Rongea* develops in the riverbed oriented E–W about 5.8 km north of Prasat Thom (Fig. 2). It has been first mentioned by a Japanese team which however mistake the river for the *O Ta Lich* (Moriai et al., 2002), and later described more in detail by Evans (Evans, 2009). Pits and channels are located in the riverbed outcrop, which consists of a massive sandstone terrace with festoon cross-bedding. Although the major marks are located in the riverbed, there is evidence that the quarry could continue underneath the rice fields which surround the river. For this reason, its real extension is difficult to estimate.

The last quarry, located E of *Prasat Trapeang Russei*, is so far the biggest known in Koh Ker, covering an area of about 6 ha (Fig. 2). First mentioned by Evans (Evans, 2009), the quarry extension has been assessed only during the recent field trip in February 2010, when was also possible to collect samples from the numerous pits and channels aligned N–S along a small tributary of *Stung Rongea* (Fig. 1). As for the other Koh Ker quarries, blocks are dressed from massive boulders as well as terraces in various sizes and shapes, without a clear systematic organization. One unfinished sculpture of a sitting lion is still lying on the quarry site, suggesting that both building stones and sculpture were produced and roughly dressed. One sample from the riverbed of the *Ta Ong* river, along the ancient road that connects Beng Mealea to Bakan (also known as Preah Khan of Kompong Svay), has been also included in the study. In total, 44 samples have been collected, as listed in detail in Tables 1 and 2.

Table 1
Selected petrographic and textural parameters of samples from Kulen Mountain quarry district.

Sample	Q^* (%)	F^* (%)	L^* (%)	Lm (%)	Lv (%)	$Ls + C$ (%)	Heavy minerals (%)	Mean (mm)	Median (mm)	Sorting (σ)	Matrix (%)	Cement (%)	Cement type	Site	Location	Type
PB3	50.3	42.0	7.7	40.9	18.2	40.9	0.7	0.14	0.15	0.59	6.1	7.2	Ch	Phnom Bei	13°30′4.30″N 104°12′39.40″E	Quarry
PB4	60.1	35.4	4.4	41.7	41.7	16.7	5.0	0.09	0.09	0.50	1.7	10.0	Ch C	Phnom Bei	13°29′59.10″N 104°12′37.20″E	Quarry
PB72	54.3	38.3	7.4	23.5	58.8	17.6	7.0	0.13	0.13	0.56	6.8	10.0	Ch	Phnom Bei	13°29′59.00″N 104°13′4.60″E	Outcrop
PB82	54.5	43.5	1.9	33.3	33.3	33.3	4.2	0.13	0.14	0.55	6.5	9.0	Ch	Phnom Bei	13°30′44.10″N 104°12′33.00″E	Outcrop
PB85	68.3	28.6	3.1	42.9	28.6	28.6	4.5	0.08	0.08	0.50	5.1	2.5	Ch C	Phnom Bei	13°30′34.60″N 104°12′26.40″E	Outcrop
TA1(T)	49.0	44.3	6.8	30.0	35.0	35.0	3.2	0.14	0.14	0.57	2.8	6.2	Ch C	Thma Andong	13°29′2.10″N 104°12′20.80″E	Quarry
TA1(B)	41.8	51.4	6.8	21.1	47.4	31.6	3.0	0.14	0.15	0.55	2.6	9.1	Ch	Thma Andong	Same as above	Quarry
TA5	50.4	44.5	5.1	42.9	21.4	35.7	2.0	0.12	0.12	0.58	3.0	6.1	Ch	Thma Andong	13°29′2.40″N 104°12′27.00″E	Quarry
TA6(T)	51.4	45.9	2.7	14.3	28.6	57.1	4.0	0.08	0.08	0.54	2.6	9.4	Ch	Thma Andong	13°29′4.20″N 104°12′27.70″E	Quarry
TA6(M)	52.9	39.9	7.2	15.8	52.6	31.6	3.0	0.10	0.09	0.57	3.2	10.1	Ch	Thma Andong	Same as above	Quarry
TA6(B1)	47.3	44.3	8.4	43.5	43.5	13.0	4.3	0.13	0.14	0.62	3.1	9.6	Ch C	Thma Andong	Same as above	Quarry
TA6(B2)	55.6	41.5	3.0	12.5	50.0	37.5	2.4	0.09	0.09	0.45	3.0	7.7	C Ch	Thma Andong	Same as above	Quarry
TTD1	54.4	41.2	4.4	58.3	16.7	25.0	4.0	0.09	0.09	0.49	3.0	10.1	Ch C	Trapeang Thmo Dap	13°29′9.40″N 104°13′19.80″E	Quarry
TTD2	57.7	40.2	2.1	16.7	50.0	33.3	4.6	0.09	0.09	0.56	2.0	6.9	Ch C	Trapeang Thmo Dap	13°29′11.60″N 104°13′21.10″E	Quarry
OTD	48.1	43.8	8.1	39.1	47.8	13.0	2.7	0.15	0.16	0.47	1.7	12.5	Ch	O ThmoDap	13°28′52.70″N 104°13′32.60″E	Quarry
TL91	53.6	43.8	2.6	50.0	33.3	16.7	9.2	0.12	0.12	0.63	5.1	8.3	Ch	Toek Lick	13°28′47.60″N 104°12′14.10″E	Quarry
TL104(T)	56.6	34.6	8.8	11.1	50.0	38.9	5.0	0.18	0.18	0.50	5.8	13.5	Ch C	Toek Lick	13°28′49.70″N 104°12′24.60″E	Quarry
TL104(M)	54.4	36.0	9.6	28.0	56.0	16.0	2.6	0.16	0.16	0.56	4.0	11.4	Ch	Toek Lick	Same as above	Quarry
TL104(B)	56.7	34.9	8.4	16.7	66.7	16.7	1.3	0.12	0.12	0.56	5.5	9.8	Ch	Toek Lick	Same as above	Quarry
TL106	58.4	35.8	5.8	42.9	35.7	21.4	4.2	0.14	0.15	0.52	3.3	18.2	Ch C	Toek Lick	13°28′53.30″N 104°12′24.50″E	Quarry
TL109(T)	55.8	37.3	6.9	16.7	50.0	33.3	5.7	0.17	0.17	0.67	1.8	12.1	Ch	Toek Lick	13°28′57.80″N 104°12′21.30″E	Quarry
TL109(M)	57.7	35.9	6.4	33.3	40.0	26.7	12.8	0.14	0.15	0.53	1.5	15.7	Ch	Toek Lick	Same as above	Quarry
TL109(B)	61.1	27.2	11.7	15.8	47.4	36.8	3.1	0.10	0.10	0.56	1.8	38.7	C	Toek Lick	Same as above	Quarry
TO132	55.5	40.6	3.9	10.0	50.0	40.0	3.0	0.15	0.16	0.59	4.1	10.7	Ch	Ta Ong	13°25′25.80″N 104°24′20.00″E	Outcrop

T: sample taken at the top of the outcrop or quarry surface; M: sample taken at the middle height of the outcrop or quarry surface; B: sample taken at the base of the outcrop or quarry surface; C: Calcite; Ch: Chlorite.

2.2. Analytical methods

Cores of fresh stone have been drilled perpendicularly to the bedding from the larger fragments collected from the quarry site. Standard thin sections have been prepared from fragments embedded in epoxy resin, perpendicularly to the lamination/bedding plane as seen on the hand sample. For each sample, the number of detrital grains was counted and each grain classified and measured by means of a micrometric eyepiece using a Zeiss Axioplan II polarized light microscope. Counts varied according to the dimension of samples and on the grain size and sorting of the rock, but usually at least 300 points have been counted (Gazzi, 1966; Dickinson, 1970). The nature of detrital minerals has been further qualitatively investigated by scanning electron microscopy and energy dispersive X-ray spectroscopy of carbon-coated thin sections (LEO 1455VP electron microscope equipped with a Link Pentafet Si(Li) SATW EDS detector).

In addition to petrographic analysis, bulk geochemical analyses have been performed on crushed samples from the same cores. Elemental composition was determined by means of instrumental neutron activation analysis (INAA) and a combination of inductively coupled plasma emission and mass spectroscopy (ICP-OES and ICP-MS) by Activation Laboratories Ltd, Canada.

3. Results and discussion

3.1. Geography and geology of the studied area

The studied area is located between the provinces of Siem Reap and Preah Vihear, and occupies a surface of about 6000 km^2 which connects the Tonle Sap plain in the south to the Dangrek range, which marks the natural boundary with Thailand towards north.

The region is characterized by loosely consolidated Quaternary sediments, locally strongly laterized and with average altitude of 80–100 m, from which numerous buttes rise to a maximum high of 500 m in the eastern sector. In the surveyed area, these buttes form a wide amphitheatre open toward north which includes, from west to east, the Kulen plateau, the hills of *Phnom Pours*, and the Tbeng plateau. The Kulen and Tbeng mountains share similar lithology, being mainly constituted of the Upper Jurassic–Cretaceous suite known in Cambodia as *Gres Superieurs* which, together with the Lower–Middle Jurassic *Terrain Rouge*, constitute the broad sub-horizontal highlands of western and northern Cambodia (Fig. 1) (United Nations, 1993).

In this framework, the areas of Beng Mealea and Koh Ker are of specific geological and archaeological interest, the former because

Table 2
Selected petrographic and textural parameters of samples from Koh Ker quarry district.

Sample	Q^* (%)	F^* (%)	L^* (%)	*Lm* (%)	*Lv* (%)	*Ls* + *C* (%)	Heavy minerals (%)	Mean (mm)	Median (mm)	Sorting (σ)	Matrix (%)	Cement (%)	Cement type	Site	Location	Type
SR4	50.0	44.1	5.9	17.6	76.5	5.9	3.3	0.19	0.19	0.60	1.1	11.0	Ch C Qz F	Stung Rongea	13°50′9.80″N 104°32′23.00″E	Quarry
SR5	47.3	46.4	6.3	14.3	71.4	14.3	3.9	0.19	0.19	0.64	2.6	10.2	Ch C Qz	Stung Rongea	13°50′8.90″N 104°32′22.90″E	Quarry
TR53	53.4	40.6	6.0	33.3	40.0	26.7	3.8	0.21	0.21	0.59	5.3	6.6	Ch	Trapeang Russei	13°49′15.10″N 104°33′43.60″E	Quarry
TR54	55.3	39.9	4.8	10.0	40.0	50.0	1.0	0.26	0.28	0.52	5.5	9.3	Ch	Trapeang Russei	13°49′17.70″N 104°33′43.60″E	Quarry
TR55	58.0	36.4	5.6	28.6	50.0	21.4	1.6	0.27	0.29	0.57	2.6	4.0	Qz Ch	Trapeang Russei	13°49′18.70″N 104°33′42.60″E	Quarry
TR56	57.2	38.0	4.8	0.0	60.0	40.0	1.4	0.26	0.29	0.63	6.8	6.4	Ch	Trapeang Russei	13°49′21.30″N 104°33′42.60″E	Quarry
TR57	65.1	29.0	6.0	13.3	40.0	46.7	0.8	0.27	0.29	0.60	4.7	14.5	Ch	Trapeang Russei	13°49′20.80″N 104°33′43.00″E	Quarry
TR58	50.2	42.9	6.9	6.3	81.3	12.5	3.7	0.23	0.22	0.61	2.3	15.9	Ch	Trapeang Russei	13°49′22.20″N 104°33′43.50″E	Quarry
TR59	54.6	39.8	5.6	35.7	35.7	28.6	1.2	0.22	0.22	0.60	4.7	14.4	Ch	Trapeang Russei	13°49′22.30″N 104°33′41.90″E	Quarry
TR62	51.5	43.7	4.8	0.0	63.6	36.4	0.8	0.19	0.19	0.55	5.9	14.9	Ch	Trapeang Russei	13°49′37.00″N 104°33′43.00″E	Quarry
TR64	54.4	38.7	6.9	14.3	64.3	21.4	4.6	0.22	0.22	0.67	3.4	14.3	Ch Qz	Trapeang Russei	13°49′34.10″N 104°33′43.10″E	Quarry
TR66	55.8	38.4	5.8	35.7	35.7	28.6	4.3	0.2	0.2	0.7	4.1	14.1	Ch	Trapeang Russei	13°49′31.20″N 104°33′43.00″E	Quarry
PT1	43.9	51.6	4.5	46.2	30.8	23.1	1.9	0.12	0.12	0.58	3.4	5.7	Ch Qz	Prasat Thneng	13°47′13.10″N 104°32′54.60″E	Quarry
PT2	46.3	50.6	3.1	37.5	37.5	25.0	6.3	0.13	0.14	0.61	2.7	10.2	Ch Qz F	Prasat Thneng	Same as above	Quarry
PTG1	49.1	46.3	4.6	46.2	46.2	7.7	3.0	0.12	0.12	0.65	3.0	4.8	Ch C	Prasat Thneng – TempleG	13°47′5.10″N 104°32′48.60″E	Carved outcrop
PTH2	56.7	39.4	4.0	27.3	63.6	9.1	2.6	0.12	0.12	0.62	4.0	5.2	Ch C	Prasat Thneng – TempleH	13°47′9.90″N 104°32′46.10″E	Carved outcrop
PTI4	57.7	38.4	3.9	27.3	63.6	9.1	2.3	0.12	0.12	0.65	5.3	4.1	Ch C	Prasat Thneng – Temple I	13°47′13.16″N 104°32′44.00″E	Carved outcrop
AK1	45.2	43.2	11.6	26.5	61.8	11.8	1.7	0.16	0.17	0.52	2.6	8.6	Ch	Ang Khna	13°45′57.10″N 104°32′55.60″E	Quarry
AK5	51.6	41.2	7.3	19.0	71.4	9.5	3.6	0.10	0.10	0.52	2.9	8.6	Ch C Qz	Ang Khna	13°46′0.70″N 104°32′52.60″E	Quarry
AK8	49.8	43.6	6.6	12.5	68.8	18.8	18.8	0.15	0.16	0.49	1.4	10.7	Ch Qz	Ang Khna	13°46′5.40″N 104°32′57.90″E	Quarry

C: Calcite; Ch: Chlorite; F: Feldspar; Qz: Quartz.

is located close to the contact between the *Gres Superieurs*, forming the steep slopes and summit of Kulen, and the *Terrain Rouge,* outcropping at its eastern foothills. The latter because is scattered with numerous outcrops of Lower–Middle Jurassic sandstone, brought to light by the dense network of small seasonal rivers which cut across the thin, mostly unconsolidated Quaternary sediment (Moriai et al., 2002).

Of the few published geological field works, that by J.P. Contri (1972) reports the geological section as seen on the south-eastern slope of Kulen Mountain (Fig. 1). Schematically, from the bottom to the top, the sub-horizontal sequence includes: 40 m of thin bedded, fine-grained feldspathic sandstone, where the Angkor quarries are located; 60 m of fine-grained, red sandstone with red clay interbeds, topped by a polygenic conglomerate; 30 m of grayish-green, fine-grained sandstone in beds of 30–50 cm. The sequence ends with 200–220 m of massive, cross bedded, medium to fine-grained quartz-rich sandstone with clay horizons. This level is characterized by quartz and chert pebbles and conglomerate levels. The sandstone color varies from white to brownish-red depending on the concentration of iron oxides (Contri, 1972).

The described sequence has been correlated to the lower and middle horizons of the Khorat sequence of Jurassic–Lower Cretaceous age (Contri, 1972; Racey et al., 1996; Meesook et al., 2002; Uchida et al., 2010). It is necessary to bear in mind that the age of such sedimentary formations is problematic, and that the evidence for a Jurassic–Lower Cretaceous age of these sequences is not conclusive (Racey et al., 1996).

Based on these correlations and on field evidences, the numerous quarries at the foothill of Kulen Mountain open on the upper portion of the *Terrain Rouge* Formation, close to the contact with the *Gres Superieurs*. This section of the stratigraphic sequence is traditionally considered of sub-continental origin, and differs from the lower portion, characterized by red conglomerate, for the abundance of red to green feldspathic sandstone (Contri, 1972; Alabouvette, 1973; United Nations, 1993; Racey et al., 1996; Sotham, 1997).

At Koh Ker no direct evidence exists to show the quarry's position within the sedimentary sequence, although the almost horizontal attitude of these formations and the presence of outcrops of *Gres Superieurs* of similar attitude few kilometers east suggest a setting analogous to Beng Mealea.

Both sites are included in the 1:200,000 *Tbeng Meanchey* sheet published by the Department of Geology, General Department of Mineral Resources of Cambodia and based on the reconnaissance mapping carried out in 1967/78 by the French *Bureau de Recherches Geologiques et Minières* (Contri, 1972).

At the foothill of Kulen Mountain the outcrop is mapped as Upper Triassic–Lower Jurassic (T_3–J_1), while more recent

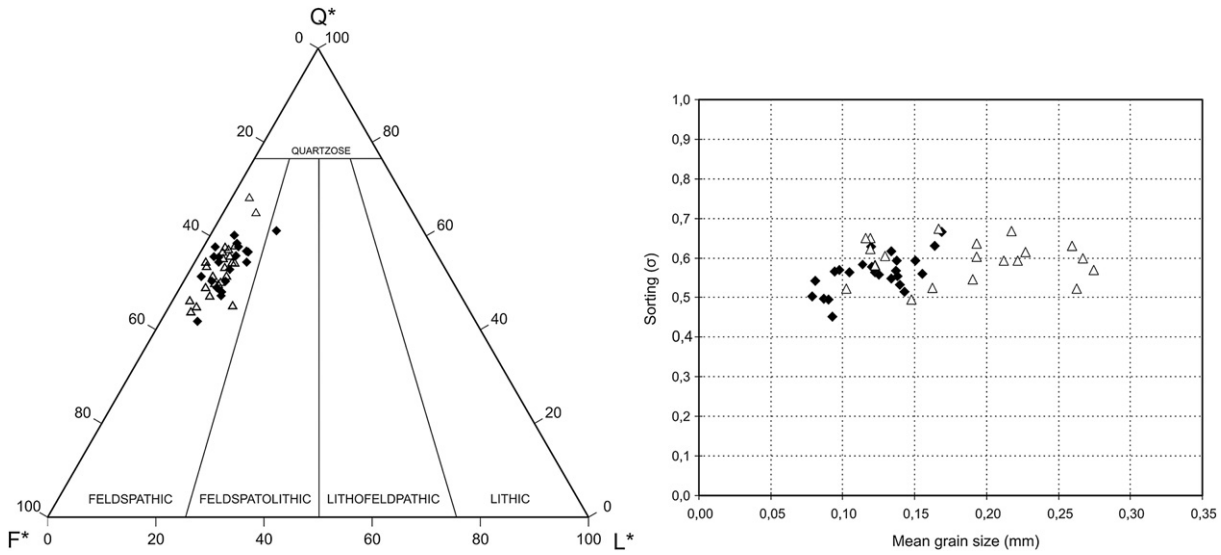


Fig. 3. Petrographic (left) and textural (right) classification of sandstone samples from the eastern foothill of Kulen Mountain (full diamonds) and from Koh Ker (void triangles).

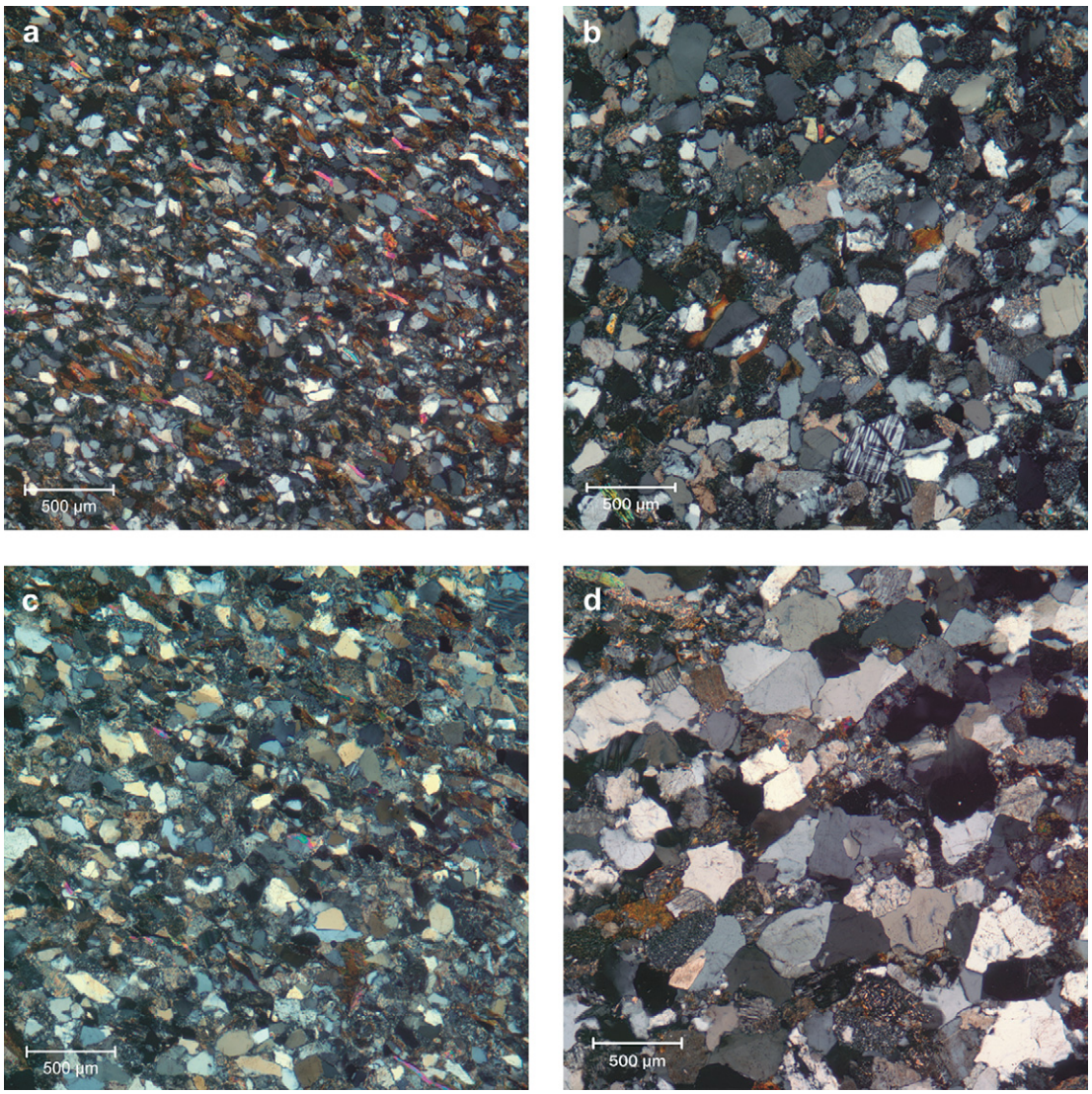


Fig. 4. Micrographs showing the two sandstone grain-size end-members from each quarry district. a–b) samples from the eastern foothill of Kulen Mountain; c–d) samples from Koh Ker. Crossed polars.

Table 3
Concentration of major elements (wt %) of samples from Kulen Mountain quarry district.

Sample	SiO_2 %	Al_2O_3 %	Fe_2O_3(T) %	MnO %	MgO %	CaO %	Na_2O %	K_2O %	TiO_2 %	P_2O_5 %	LOI %	Σ %
PB3	71.70	12.76	3.88	0.05	1.40	0.69	2.66	2.33	0.35	0.10	3.02	98.92
PB4	68.10	11.68	5.13	0.08	2.23	3.64	2.26	1.93	0.73	0.21	4.79	100.80
PB72	74.25	11.70	3.51	0.04	1.31	0.97	2.60	1.33	0.55	0.16	3.43	99.84
PB82	72.65	12.79	3.80	0.06	1.70	0.95	2.90	1.74	0.52	0.12	3.26	100.50
PB85	70.61	11.23	3.50	0.06	1.70	2.62	2.48	1.56	0.52	0.13	4.28	98.70
TA1(T)	69.71	12.39	5.45	0.10	2.07	1.28	2.90	2.13	0.57	0.17	2.99	99.76
TA1(B)	71.02	11.98	5.40	0.07	2.03	0.90	2.95	2.14	0.55	0.12	2.79	99.97
TA5	70.36	11.89	5.04	0.07	2.06	0.96	2.77	2.07	0.59	0.14	2.80	98.75
TA6(T)	69.70	13.11	5.46	0.09	2.15	0.97	2.71	2.40	0.58	0.16	3.28	100.60
TA6(M)	70.00	12.31	3.66	0.06	1.55	2.16	2.72	2.57	0.58	0.14	3.74	99.49
TA6(B1)	69.57	12.50	5.60	0.08	2.28	1.64	2.83	2.11	0.75	0.17	3.08	100.60
TA6(B2)	67.66	11.73	3.70	0.07	1.60	3.44	2.69	2.47	0.54	0.14	4.64	98.66
TTD1	69.12	12.08	5.31	0.07	2.61	2.42	2.23	1.74	0.65	0.18	4.44	100.90
TTD2	68.18	11.88	5.09	0.08	2.48	3.18	2.25	1.73	0.59	0.15	4.93	100.50
OTD	70.82	12.73	4.44	0.06	2.27	0.52	2.85	1.98	0.34	0.10	2.95	99.06
TL91	68.97	12.30	5.62	0.13	2.18	1.14	2.89	2.09	0.78	0.20	2.89	99.19
TL104(T)	71.30	12.23	4.45	0.07	1.91	1.65	3.16	2.05	0.56	0.10	3.15	100.60
TL104(M)	71.32	12.36	4.50	0.08	1.93	1.02	3.12	2.09	0.61	0.14	2.68	99.84
TL104(B)	70.01	13.42	4.75	0.08	2.43	0.88	3.03	2.39	0.56	0.14	3.12	100.80
TL106	68.88	12.90	4.75	0.08	2.31	1.58	3.05	2.12	0.65	0.14	3.14	99.60
TL109(T)	72.11	11.93	4.38	0.08	1.76	1.32	3.21	2.00	0.61	0.15	2.61	100.20
TL109(M)	69.31	12.81	5.40	0.07	1.51	1.88	3.06	1.88	1.08	0.27	3.15	100.40
TL109(B)	46.58	8.53	2.12	0.47	1.13	19.79	2.08	1.77	0.37	0.11	17.01	99.95
TO132	72.02	12.49	3.97	0.06	1.85	0.70	2.77	1.40	0.40	0.11	3.99	99.76

T: sample taken at the top of the outcrop or quarry surface; M: sample taken at the middle height of the outcrop or quarry surface; B: sample taken at the base of the outcrop or quarry surface.

reviews assign it to Lower to Middle Jurassic (J_{1-2}) (United Nations, 1993; Sotham, 1997). On the contrary, no outcrops are mapped in the vicinity of Koh Ker, where the sandstone occupies discontinuously riverbeds and low terraces of difficult identification.

3.2. Petrography of the quarried sandstones

The studied sandstone samples are characterized by similar composition and texture. Variations in structure, texture and, to a lesser extent in mineral content, do occur at the scale of the single pit or outcrop in the studied quarry sites, and are related to local changes in the patterns of sedimentation. All the analyzed samples are feldspathic arenite of average composition $Q_{54}F_{40}L_6$ (Fig. 3).

The sandstone is composed of very fine to fine, moderately well sorted, sub-angular to rounded grains cemented by predominant authigenic chlorite (Fig. 4). Cement averages 10.4%, although the degree of compaction and cementation can vary in the single outcrop. Calcite cement, authigenic quartz and feldspar can be also present in variable amounts, with the latter two being more abundant in sandstone from Koh Ker.

Mono-(Q_m), poly- and microcrystalline (Q_p) quartz constitute more than half of the framework grains ($Q^* = 53.6 \pm 5.3\%$), with $Q_m > Q_p$. Undulatory quartz ($23.7 \pm 3.7\%$) prevails over quartz with non-undulose extinction ($17.9 \pm 4.7\%$).

Table 4
Concentration of major elements (wt %) of samples from Koh Ker quarry district.

Sample	SiO_2 %	Al_2O_3 %	Fe_2O_3(T) %	MnO %	MgO %	CaO %	Na_2O %	K_2O %	TiO_2 %	P_2O_5 %	LOI %	Σ %
SR4	71.95	10.79	3.90	0.07	1.34	2.27	2.62	1.98	0.35	0.10	3.11	98.49
SR5	69.63	11.23	4.37	0.09	1.62	3.06	2.60	2.04	0.37	0.11	4.00	99.12
TR53	76.80	11.23	2.65	0.04	1.08	0.69	2.79	1.26	0.38	0.08	2.90	99.90
TR54	77.82	10.28	2.75	0.04	1.62	0.91	2.77	1.22	0.22	0.06	2.49	100.20
TR55	82.86	8.16	1.27	0.06	0.51	1.08	2.65	1.00	0.14	0.16	1.87	99.77
TR56	79.54	10.01	2.35	0.04	0.81	0.62	2.54	1.62	0.18	0.06	2.26	100.00
TR58	76.51	10.80	3.20	0.05	1.38	0.74	2.85	1.45	0.30	0.07	2.49	99.84
TR59	76.32	10.97	3.32	0.05	1.52	0.81	2.97	1.32	0.31	0.07	2.54	100.20
TR62	74.10	11.93	3.72	0.06	1.85	0.80	2.94	1.56	0.29	0.07	2.95	100.30
TR64	77.40	10.53	3.23	0.05	1.58	0.80	3.07	1.08	0.36	0.09	2.40	100.60
TR66	74.37	11.85	3.58	0.05	1.52	0.93	3.23	1.19	0.44	0.10	3.11	100.40
PT1	71.18	12.24	4.86	0.06	1.56	0.83	2.91	1.87	0.43	0.13	2.87	98.94
PT2	70.11	12.60	4.87	0.11	1.64	1.10	2.87	1.87	0.55	0.14	3.25	99.09
PTG1	70.33	11.93	3.81	0.06	1.36	1.59	2.91	2.08	0.53	0.13	3.94	98.68
PTH2	68.43	13.34	5.17	0.08	1.93	0.96	2.65	2.01	0.62	0.16	3.73	99.09
PTI4	68.32	13.31	4.13	0.07	1.74	1.74	2.59	2.15	0.60	0.17	4.57	99.39
AK1	73.62	11.67	3.23	0.05	1.26	1.18	2.86	1.77	0.26	0.09	3.16	99.14
AK5	77.56	8.94	3.06	0.05	0.86	1.95	2.29	1.14	0.34	0.09	2.91	99.18
AK8	59.10	12.32	10.05	0.22	2.08	5.09	2.18	1.02	2.56	0.34	4.66	99.62

Table 5
Concentration of selected minor and trace elements (in parts-per-million, ppm) of samples from Kulen Mountain quarry district.

	Be ppm	Sc ppm	V ppm	Cr ppm	Co ppm	Ni ppm	Cu ppm	Zn ppm	Rb ppm	Sr ppm	Y ppm	Zr ppm	Ba ppm	Hf ppm	La ppm	Ce ppm	Nd ppm	Sm ppm	Eu ppm	Yb ppm	Pb ppm	Th ppm	U ppm
PB3	1.5	6.8	50.0	42	11	27	9	54	60	181	12.1	110	510	3.5	23.2	54	18	3	0.9	1.5	7	7.1	1.4
PB4	1.6	12.0	77.3	77	11	31	10	65	60	180	25.9	432	380	10.3	51.0	95	35	6.3	1.8	3.1	11	15.6	3.0
PB72	2.0	8.7	60.0	45	7	30	10	52	20	196	20.0	176	396	4.1	31.6	50	21	4.5	0.7	1.4	6	6.8	1.6
PB82	1.0	10.7	73.0	52	8	24	23	57	20	182	18.0	212	704	4.6	37.7	62	22	5	1.0	1.7	11	8.3	2.2
PB85	1.0	7.3	56.0	44	8	28	11	46	30	159	20.0	186	318	4.2	31.2	52	23	4.6	1.0	1.7	9	6.5	2.0
TA1(T)	1.6	10.3	66.4	64	12	30	8	51	80	193	19.7	235	500	6.8	43.1	74	31	5.3	1.6	2.3	8	11.8	2.7
TA1(B)	1.8	10.4	63.9	61	13	30	6	52	50	190	17.7	245	630	6.9	39.4	76	28	5	1.5	2.3	5	11.1	2.1
TA5	1.8	10.8	69.2	62	12	31	11	72	60	187	19.9	316	470	7.7	44.7	85	36	5.4	1.5	2.5	8	12.3	3.0
TA6(T)	1.9	10.4	68.0	58	15	38	82	74	70	190	18.2	230	580	5.8	30.8	66	21	4.4	1.4	2.3	8	11.3	1.6
TA6(M)	2.2	11.0	80.8	61	12	31	17	57	70	192	18.9	263	540	6.2	37.1	70	29	4.9	1.4	2.4	10	11.5	3.3
TA6(B1)	1.6	12.1	84.0	78	12	35	15	75	60	204	23.6	435	570	10.6	53.7	103	38	6.4	1.7	2.8	8	15.6	2.1
TA6(B2)	2.0	10.0	57.6	57	12	31	31	53	80	188	18.6	261	450	6.5	38.9	72	32	5.1	1.5	2.4	8	11.5	2.8
TTD1	1.7	10.8	68.5	65	14	36	10	65	70	155	20.2	245	710	6.5	38.7	73	27	5.1	1.4	2	7	11.9	1.4
TTD2	1.6	11.4	73.8	67	12	33	11	70	10	153	22.4	317	450	8.1	45.8	82	38	5.5	1.7	2.8	6	13.8	2.8
OTD	1.6	8.2	52.3	40	10	28	12	65	70	163	13.5	99	830	3.2	32.6	52	22	3.6	1.2	1.3	4	7.6	1.7
TL91	2.0	7.5	70.0	48	14	33	21	63	50	200	20.0	201	501	4.2	30.2	53	17	4.4	1.0	1.7	7	7.0	1.8
TL104(T)	1.0	9.2	88.0	57	11	31	4	66	30	205	25.0	486	495	9.6	47.8	81	27	6.2	1.2	2.3	11	12.1	2.5
TL104(M)	2.0	7.0	66.0	42	11	24	14	53	30	217	23.0	238	866	4.7	42.3	64	23	4.9	1.2	1.7	11	7.4	2.7
TL104(B)	2.0	7.9	71.0	47	11	24	6	50	20	216	21.0	294	721	6.0	37.8	65	26	5.5	1.2	1.8	8	9.0	2.3
TL106	2.0	8.8	81.0	49	11	28	4	55	20	211	22.0	319	442	6.9	41.9	74	29	5.3	1.2	2	7	8.5	3.1
TL109(T)	2.0	7.5	67.0	48	8	21	8	51	40	213	24.0	313	694	6.2	40.8	72	23	5.6	1.3	1.8	9	8.8	2.3
TL109(M)	1.0	13.3	99.0	83	10	29	7	67	70	259	34.0	294	495	6.4	67.3	129	47	9.5	2.0	2.9	9	12.8	3.6
TL109(B)	1.0	5.6	44.0	31	8	20	8	31	50	188	30.0	130	374	2.6	31.3	52	25	4.8	1.0	2.2	9	5.9	2.3
TO132	2.0	5.5	46.0	35	7	21	7	48	60	157	14.0	148	316	3.1	24.7	42	16	3.4	0.9	1.3	5	5.4	1.8

T: sample taken at the top of the outcrop or quarry surface; M: sample taken at the middle height of the outcrop or quarry surface; B: sample taken at the base of the outcrop or quarry surface.

Table 6
Concentration of selected minor and trace elements (in parts-per-million, ppm) of samples from Koh Ker quarry district.

	Be ppm	Sc ppm	V ppm	Cr ppm	Co ppm	Ni ppm	Cu ppm	Zn ppm	Rb ppm	Sr ppm	Y ppm	Zr ppm	Ba ppm	Hf ppm	La ppm	Ce ppm	Nd ppm	Sm ppm	Eu ppm	Yb ppm	Pb ppm	Th ppm	U ppm
SR4	1.2	6.7	39.1	34	6	17	9	39	70	190.6	27.5	160.1	490	4.2	38.8	70	27	4.5	1.5	1.9	13	8.6	1.8
SR5	1.5	7.6	44.8	38	8	20	10	46	80	191.2	17.9	160.3	580	4.6	40.8	76	33	4.5	1.4	2	15	9.8	2.4
TR53	1.0	7.4	46.0	41	7	19	14	39	20	186.0	15.0	212	230	3.6	30.9	46	18	4.2	1.0	1.5	17	6.3	<0.5
TR54	1.0	5.4	31.0	28	7	14	78	40	20	197.0	31.0	78	1070	1.8	40.6	56	20	4.6	1.1	1.1	16	4.1	<0.5
TR55	1.0	3.3	20.0	15	4	9	14	17	40	176.0	14.0	77	1850	2.0	23.1	41	18	3.2	0.7	1.1	33	3.1	<0.5
TR56	2.0	4.1	26.0	15	4	16	39	36	60	176.0	29.0	90	411	2.0	87.1	34	46	9.7	2.2	2.5	25	4.1	<0.5
TR58	2.0	5.3	34.0	34	7	17	11	31	20	195.0	19.0	111	357	2.8	45.8	76	29	5.6	1.3	1.5	14	5.2	<0.5
TR59	1.0	5.6	36.0	27	6	15	31	45	20	205.0	19.0	112	368	2.7	38.1	67	21	5	1.3	1.4	14	4.9	<0.5
TR62	1.0	6.3	40.0	32	8	21	10	42	20	206.0	18.0	90	433	2.5	38.5	59	25	4.6	1.1	1.5	17	4.9	1.6
TR64	1.0	6.3	43.0	31	7	17	14	36	50	191.0	17.0	146	255	3.1	35	63	24	4.8	1.0	1.5	16	5.8	2.9
TR66	2.0	7.8	51.0	45	8	22	11	42	20	210.0	18.0	228	328	4.9	32.9	67	20	4.6	1.3	1.5	19	7.7	2.0
PT1	1.7	8.5	51.0	44	12	27	18	54	60	184.9	17.5	175.4	530	4.4	44.7	89	36	4.8	1.4	2	17	9.2	2.1
PT2	1.9	9.8	59.8	52	16	30	16	64	50	205.7	21.0	264.8	600	6.9	46.8	89	33	5.5	1.8	2.7	19	11.9	2.0
PTG1	1.9	9.2	61.1	49	10	25	19	59	50	158.5	21.0	364.1	650	8.1	44.8	86	31	5.2	1.5	2.7	61	12.6	3.1
PTH2	2.1	10.9	68.8	59	14	31	16	76	70	170.0	20.7	317.5	460	8.1	44.1	86	31	5.2	1.6	2.7	25	14.8	2.2
PTI4	2.1	10.4	67.8	70	13	39	16	52	80	159.1	20.4	298.8	460	7.4	41.4	79	31	4.9	1.4	2.6	37	13.3	2.7
AK1	6.7	6.5	34.7	69	9	25	18	46	10	280.1	370.2	88.89	1200	3.0	3060	63	1710	220	65.4	27.1	14	9.0	<0.5
AK5	<1	6.4	45.6	28	7	14	20	39	50	176.3	18.7	114.5	1500	3.6	30.4	49	21	4	1.2	1.9	10	7.2	1.7
AK8	1.7	25.1	186.7	160	13	26	26	103	10	250.2	71.3	1778	300	36.9	172	298	143	18.4	4.1	8.3	29	53.7	7.8

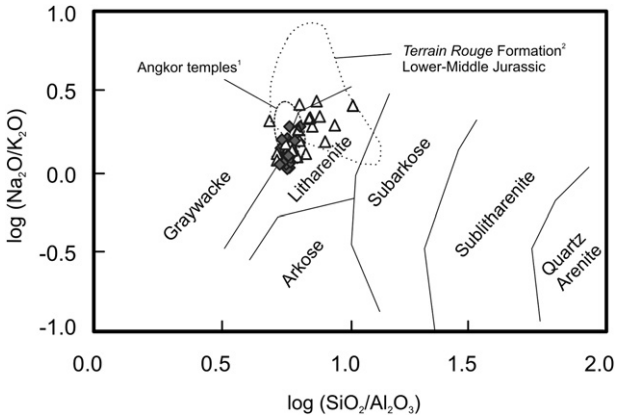


Fig. 5. Chemical classification of samples from Kulen Mountain (full diamonds) and from Koh Ker (void triangles) compared to analogous sandstone belonging to the Lower–Middle Jurassic Terrain Rouge Formation. Data are compared to (1) Kučera et al., 2008 and (2) Uchida et al., 2010.

Feldspars are predominant untwinned varieties of plagioclase and alkali feldspar (33.5 ± 4.7%), and minor polysynthetic twinned plagioclase (4.0 ± 1.6%). Both fresh and altered feldspars are present.

The abundance of rock fragments averages 5.9%, and rarely exceeds 10%. Characteristics are volcanic rocks (mostly andesite) of variable dimensions with, in order of abundance, microlithic, aphanitic and porphiric texture, as well as fragments of phyllite, quartzite, chert, micaceous schist, argillite, shale and siltstone. Rock fragments are often squashed between grains and occupy a pseudomatrix position.

Micas are minor constituents (3.1 ± 2.8%), with biotite prevailing over muscovite, and are usually bent between the framework grains. Where the sandstone is finely laminated, micas are more abundant (up to 12.4% in samples from Kulen Mountain) and parallel to the bedding planes. Biotite is often degraded or replaced by chlorite.

The heavy minerals assemblage varies from less than 1% to up to 18% of the total framework volume, as they can be concentrated in particular beds or horizons within the sequence. They consist of magnetite, hematite, epidote, ilmenite, titanite, garnet, apatite, zircon, rutile, and less abundant allanite, monazite, barite, xenotime, chromite, and tourmaline in variable relative proportion.

The overall mineral assemblage reflects a mixed provenance of felsic igneous and metamorphic rocks, with minor volcanic component (Racey et al., 1996; Kučera et al., 2008).

While the sorting of the analyzed samples is very consistent and typical of moderately well sorted sediments, grain size varies between 0.05 and 0.3 mm, i.e. from very fine- to medium-grained sandstone (Fig. 3). Grain size variations occur at the scale of the outcrop or quarry pit without specific indication of gradation. A difference in grain size is evident between the two sites, with very fine to fine sandstones concentrated in the area of Kulen Mountain (average grain size = 0.13 ± 0.03), and fine- to medium-grained in Koh Ker (average grain size = 0.19 ± 0.06 mm) (Fig. 3). The complete petrographic and textural parameters are presented in Tables 1 and 2.

3.3. Geochemistry of the quarried sandstones

Tables 3 and 4 summarize the concentration of major elements of samples from the two studied sites, while Tables 5 and 6 present the concentration of selected minor and trace elements.

Most of the samples can be chemically classified as litharenite (Herron, 1988) (Fig. 5). The data show that the sandstone has homogeneous chemical characteristics. In particular, the pit quarries located at the foothill of Kulen Mountain are characterized by a stronger homogeneity then those scattered in the Koh Ker area (Fig. 5). The high Na_2O/K_2O ratios of this sandstone suggest that most of the detrital untwinned feldspar seen in thin section is Na-rich plagioclase.

The major element abundance shows a strong chemical affinity with other analogous sandstone belonging to the Lower–Middle Jurassic *Terrain Rouge* Formation (Fig. 5). Distribution of minor and trace elements also reflects the homogeneity of this sedimentary rock. Other than rare outliers (e.g. samples AK8, extremely rich in heavy minerals, and AK1, characterized by abundant secondary precipitate), no straightforward differences have been found among the different quarries in Kulen Mountain and Koh Ker (Tables 5 and 6). A general, strong positive correlation exists among concentrations of trace elements in both sites, which can be related to the abundance of accessory minerals (Figs. 6 and 7). The geochemical compositions shown in Tables 5 and 6 are consistent with the

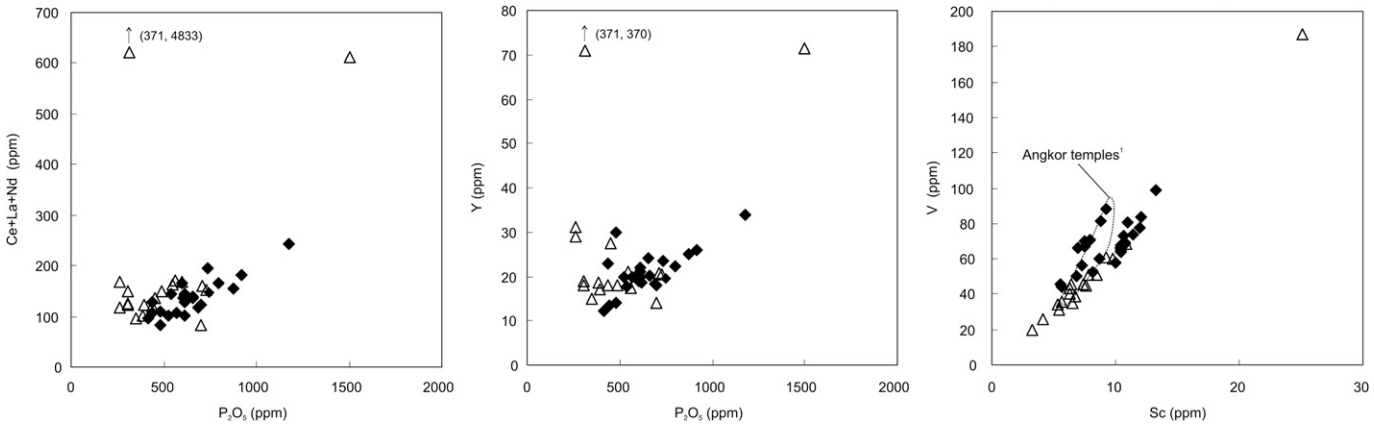


Fig. 6. Concentrations of selected trace elements in samples from Kulen Mountain (full diamonds) and from Koh Ker (void triangles). Data are compared to (1) Kučera et al., 2008.

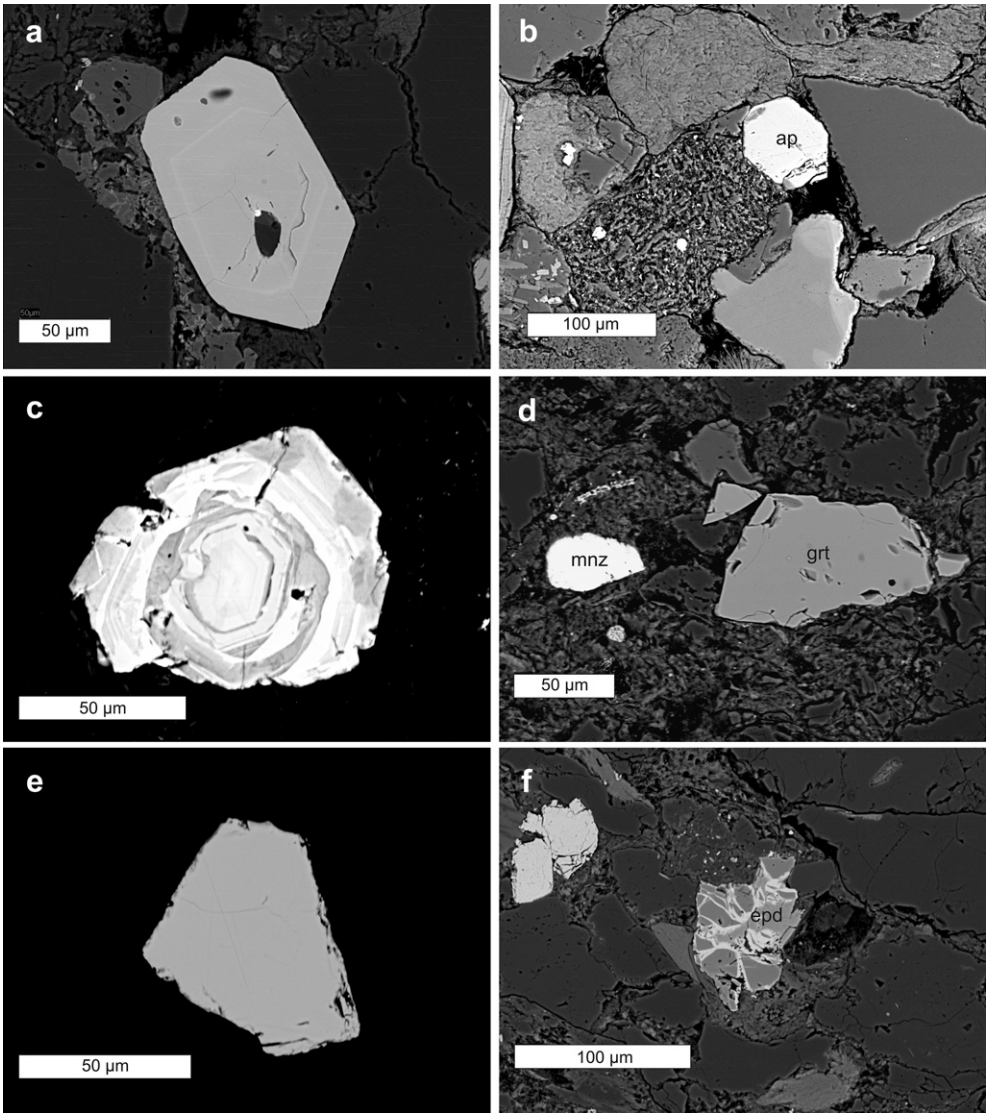


Fig. 7. Back-scattered electron (BSE) images of selected heavy minerals. a: zoned zircon from sample AK8; b: apatite (ap) from sample TA1B; c: allanite grain from sample TA1B; d: monazite (mnz) and garnet (grt) from sample PTH2; e: xenotime grain from sample 104B; f: fractured epidote (epd) with seams of LREE-epidote (bright areas) from Koh Ker samples PTH2.

concentration of minor and trace constituents of stone material from Angkor temples published by other authors (Kučera et al., 2008).

Despite the evident chemical homogeneity, variations in the concentration of specific trace element were found that might reflect a significant difference in concentration of heavy minerals among the studied samples. For instance, abundance of Light Rare Earth Elements (LREE), such as Ce, La and Nd can be related to the concentration of fractured epidote rimmed by LREE-epidote seam (Fig. 7), which is particularly abundant in Koh Ker samples. Allanite and REE-phosphate minerals, such as monazite and xenotime (Fig. 7), also account for the presence of LREE, as well as phosphorous (P), particularly in sandstone from Kulen Mountain, where the positive correlation between these elements is more marked (Fig. 6).

A significant positive correlation exists between Vanadium (V) and Scandium (Sc) (Fig. 6). These elements are mostly incorporated in ferromagnesian silicate minerals, such as biotite, as well as Fe- and Ti-oxides, such as magnetite and ilmenite, and, to a lesser extent, in phosphates and garnet (Bhatia and Crook, 1986). In this study, samples with higher V and Sc content chiefly originate from the quarries located at the foothill of Kulen Mountain, while higher V/Sc ratios are characteristic of the site of Toek Lick (Table 5). A similar positive correlation between V and Sc has been found in sandstone from Angkor temples (Fig. 6) (Kučera et al., 2008).

Finally, it is worth noticing the systematic enrichment in lead (Pb) that characterizes the samples from Koh Ker (Fig. 8).

The presence of Pb in sedimentary rocks is controlled by minerals such as feldspar, mica, sulphides, clay minerals, as well as by organic matter (Heinrichs et al., 1980). Lead is also incorporated in epidote-group minerals, zircon, and titanite (Gieré and Sorensen, 2004). However, little correlation exists among Pb and most of the major and trace elements in sandstone from Koh Ker (Fig. 8), suggesting that contamination of the samples by groundwater solutions, as well as presence of secondary mineral precipitates, could be possible sources of Pb.

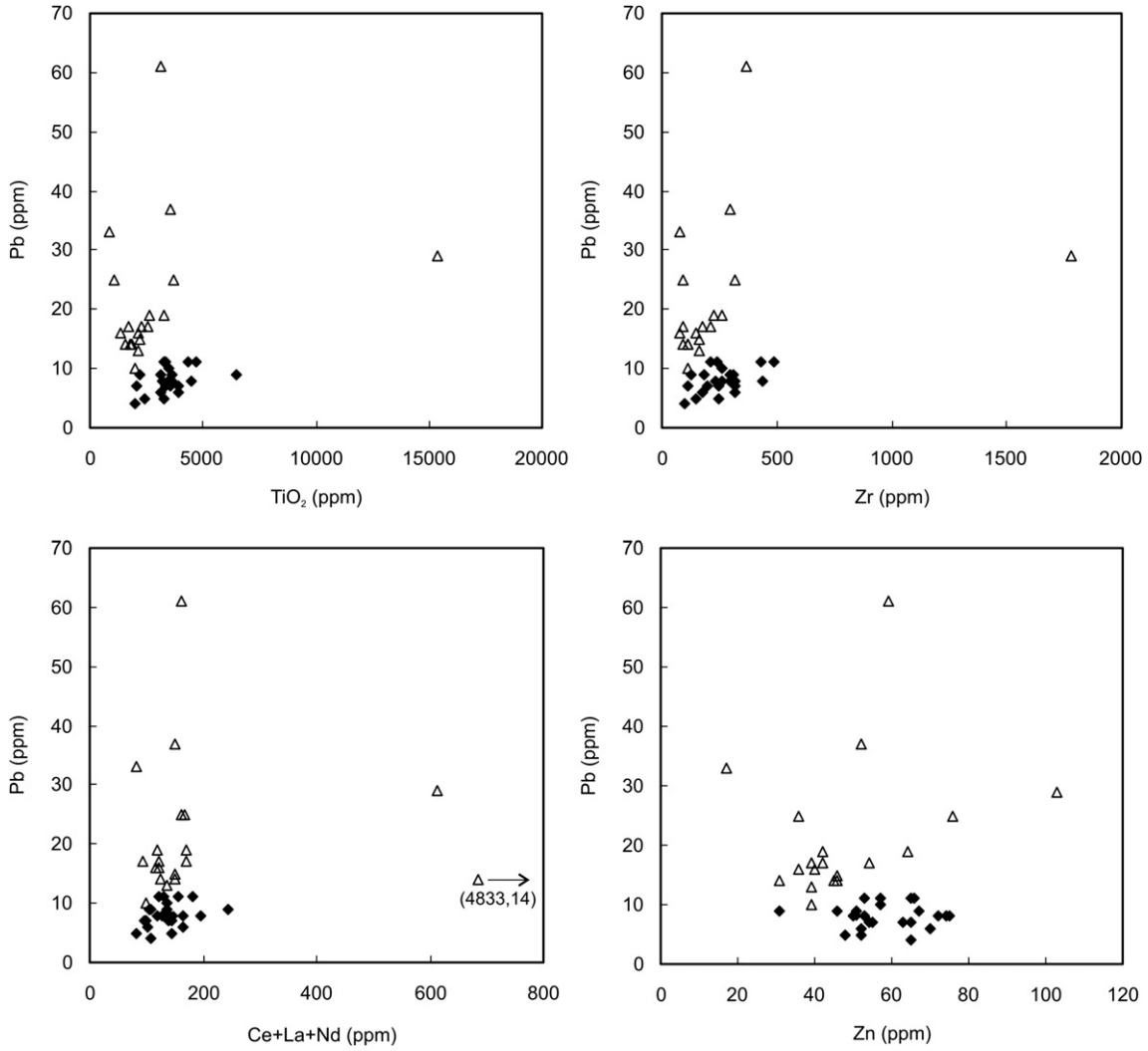


Fig. 8. Correlation between Pb content and selected minor and trace elements in samples from Kulen Mountain (full diamonds) and from Koh Ker (void triangles).

4. Conclusions

Several quarry sites and outcrops located at different heights along the eastern foothill of Kulen Mountain and in Koh Ker have been systematically sampled and characterized by means of petrographic analysis, scanning electron microscopy, and geochemical analysis. The collected samples originate from two vast areas where sandstone quarries and natural outcrops have been documented since the early archaeological exploration of Cambodia. East of Kulen Mountain, sub-horizontal sandstone beds constitute the low hills at the base of the steep mountain slope. In Koh Ker, the sandstone outcrops form vast terraces and isolated boulders often located close to, or in riverbeds.

Chisel marks, wedge holes, and channels of similar geometry are visible in both the studied areas. At the foothill of Kulen Mountain, however, the quarrying activity seems more organized and systematic, and left numerous localized pits with high stepped surfaces forming a complex network of stone exploitation. In Koh Ker, the surveyed quarries are never as extensive and developed as in the Kulen area, and the quarrying remains indicate the use of a more superficial channeling method and diffuse boulder splitting, as already pointed out by Evans (Evans, 2009).

All the studied samples have similar petrographic and chemical characteristics, and no straightforward differences have been detected within and between the two quarry sites. The sandstone is classified as very fine to fine, feldspathic arenite (Gazzi, 1966; Dickinson, 1970). Geochemically, the sandstone is classified as immature litharenite (Herron, 1988).

The overall characters are consistent with the upper portion of the Lower–Middle Jurassic *Terrain Rouge* Formation (Racey et al., 1996; Sotham, 1997; Uchida et al., 2010). This formation, which is known in Thailand under the name of Phu Kradung Formation, is characterized by immature to mature sediments mainly of lacustine and fluvial origin. Very fine- to fine-grained sandstone in massive beds is dominant in the upper part of the sequence, where it is intercalated with mudstone, siltstone and caliche horizons (Racey et al., 1996). At the foothill of Kulen Mountain, where the sequence is well exposed, it is possible to observe the different large-scale features of this sequence, including massive beds of various thickness (varying from few centimeters to a meter), but also planar-, cross-, and festoon cross-lamination. In Koh Ker, where outcrops are generally low and discontinuous, sedimentary structures are less evident, and sandstone beds massive or poorly laminated.

Within the Kulen area, the petrographic and chemical composition of the *Terrain Rouge* sandstone is very similar, and comparable to the sandstone quarried in Koh Ker. As a consequence, it is difficult to discriminate within the several existing quarries at each site, as well as between the two studied sites, located about 50 km apart.

The collected data confirm the affinity between the stone used for the construction of the Angkor temples and the litotypes outcropping at the eastern foothill of Kulen Mountain (Kučera et al., 2008; Uchida et al., 1998). However, the results also prove that it might be difficult to distinguish different building phases of Angkor temples using stone characteristics, at least by means of basic petrographic and geochemical analysis, as they are representative of vast portions of the sedimentary sequence. Similarly, the difference in grain-size between Kulen Mountain and Koh Ker cannot be safely used as a discrimination criterion, as similar grain-size variations have been registered at the outcrop scale in both sites.

A critical approach which combines petrographic analysis on major and minor constituents with major- and trace-element analysis might help in identifying different sedimentary formations, and in discriminating among quarries opened in a very homogeneous unit such as the Lower–Middle Jurassic *Terrain Rouge* Formation.

In particular, further research is needed to detail type and abundance of authigenic phases, as well as assemblage of accessory minerals. Lead isotope analysis will be useful to better understand the sources contributing to the total Pb content of the sandstone, which differs between the two studied quarry districts. Through these approaches it may become possible to assess the validity and reproducibility of petrographic and geochemical discrimination criteria in provenance studies of Angkor stone materials.

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